Dealing with the non-linearity of the material model in parametric Proper Generalized Decomposition

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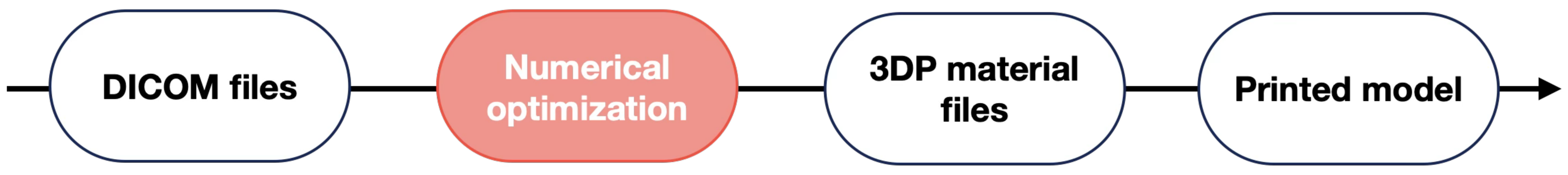
Abstract. This PhD, a part of the RHU-EndoVx project, is carried out in collaboration with the Hospital Marie Lannelongue (91). Its primary objective is to create realistic synthetic 3D-printed models of abdominal aortic aneurysms (AAA) for surgical training and rehearsals. The work involves the development of an optimization algorithm using Proper Generalized Decomposition considering hyperelastic material models to match the properties of the biomimetic materials used in the construction of the models, using the position of the load exerted by the surgeon as an extra-coordinate in the PGD decomposition.

Overarching objective

3D printing of anatomies to help surgeons train and learn the operation:
→ creating a numerical model for parametric optimization of **biomimetic** anatomies using hyperelastic modelling and PGD.



2. Incremental formulation

Loading increments to deal with NL of the model: $\boldsymbol{u}_{i+1} = \boldsymbol{u}_i + \Delta\boldsymbol{u}$

$$\mathcal{K}(\Delta\boldsymbol{u}^*, \Delta\boldsymbol{u}) = \mathcal{F}(\Delta\boldsymbol{u}^*) - \mathcal{G}(\Delta\boldsymbol{u}^*) \tag{3}$$

with $\mathcal{F}$ the loading term and $\mathcal{G}$ the term from the previous increment

→ PGD approximation: $\Delta\boldsymbol{u} \simeq \Delta\boldsymbol{u}^N = \sum_{m=1}^{N} \boldsymbol{\varphi}^m \circ \boldsymbol{\lambda}^m$

Issue: $\mathcal{K}$ has terms of order 1, 2 and 3 in $\Delta\boldsymbol{u}$.

3. Linearisation strategy

How to deal with the higher order terms for mode $\Delta\boldsymbol{u}^{n(n\leq N)}$?

Introduction of $\boldsymbol{w}$, a known approximation for $\Delta\boldsymbol{u}^n$.

→ For order 2: $\nabla^\intercal(\Delta\boldsymbol{u}^n)\nabla(\Delta\boldsymbol{u}^n) \simeq \nabla^\intercal\boldsymbol{w}\nabla(\Delta\boldsymbol{u}^n)$

Taking also into account $\Delta\boldsymbol{u}^n = \Delta\boldsymbol{u}^{n-1} + \boldsymbol{\varphi}^n \circ \boldsymbol{\lambda}^n$

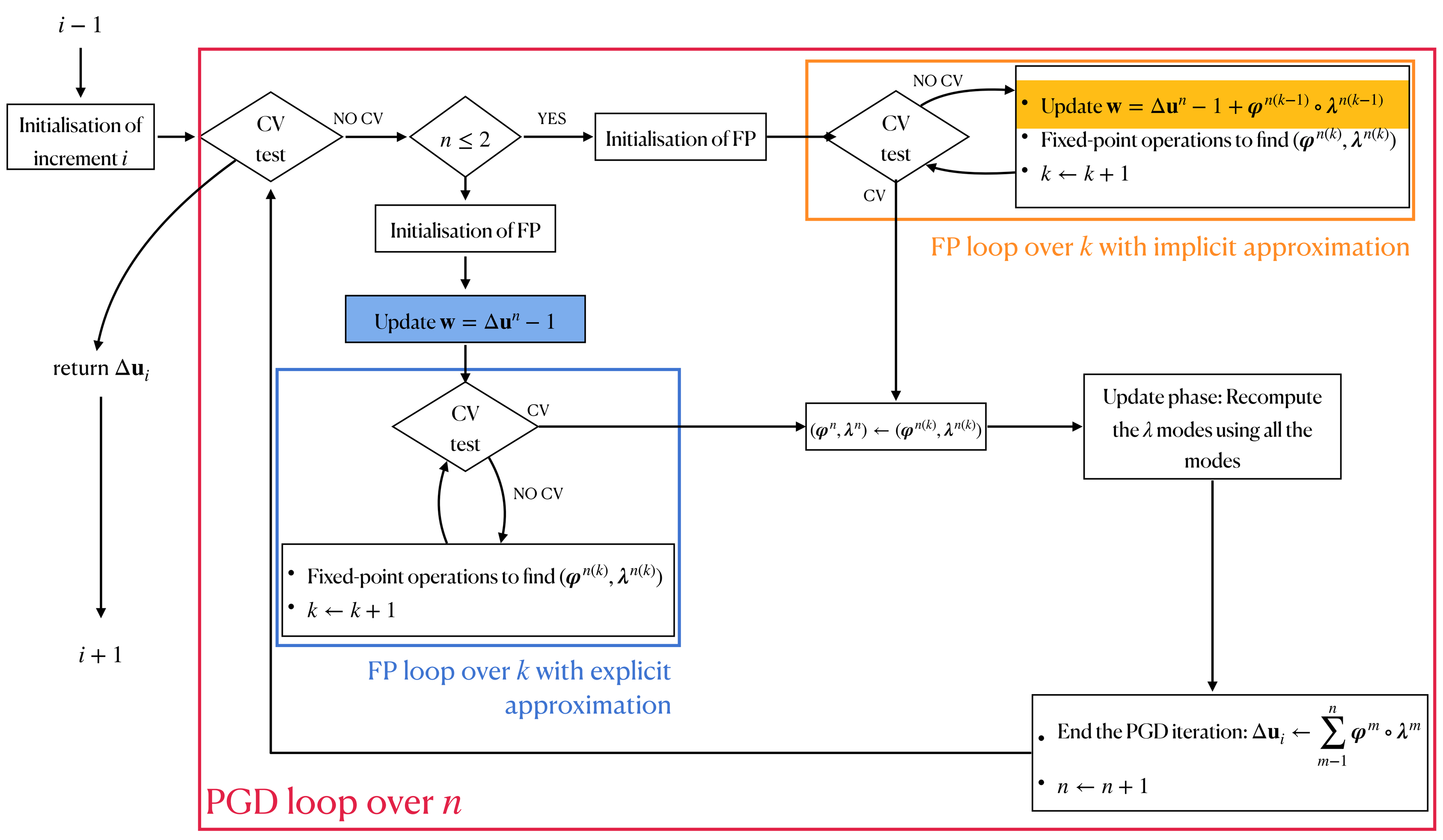
$$\mathcal{K}(\boldsymbol{w}; \Delta\boldsymbol{u}^*, \boldsymbol{\varphi}^n \circ \boldsymbol{\lambda}^n) = \mathcal{F}(\Delta\boldsymbol{u}^*) - \mathcal{G}(\Delta\boldsymbol{u}^*) - \mathcal{K}(\boldsymbol{w}; \Delta\boldsymbol{u}^*, \Delta\boldsymbol{u}^{n-1}) \tag{4}$$

What value for $\boldsymbol{w}$?

Explicit $\boldsymbol{w} = \Delta\boldsymbol{u}^{n-1}$ or implicit $\boldsymbol{w} = \Delta\boldsymbol{u}^{n-1} + \boldsymbol{\varphi}^{n(k-1)} \circ \boldsymbol{\lambda}^{n(k-1)}$

3 strategies:

full explicit (*SVK-E*), **full implicit** (*SVK-I*) or **mixed** (*SVK-M*): implicit on some modes (e.g. the first 2), explicit on the other modes



1. Mechanical problem

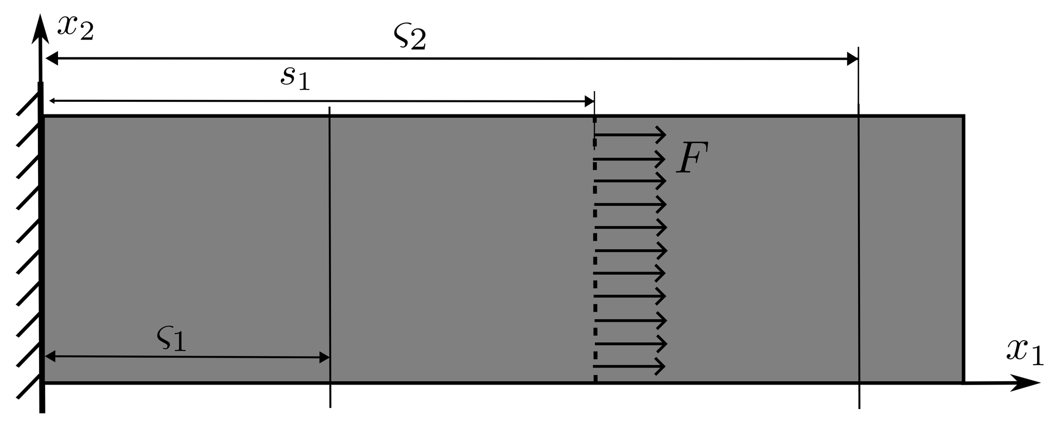


Figure 2: Sketch of the problem considered ($\mathcal{U} = \{\boldsymbol{u} \in H^1(\Omega) \backslash \boldsymbol{u} = \bar{\boldsymbol{u}}$ on $\partial_u\Omega\}$)

Saint-Venant-Kirchhoff model

$$\psi(\boldsymbol{E}) = \frac{\lambda}{2}\operatorname{tr}(\boldsymbol{E})^2 + \mu\boldsymbol{E}:\boldsymbol{E} \quad ; \quad \boldsymbol{S} = \frac{\partial\psi}{\partial\boldsymbol{E}} \tag{1}$$

PGD decomposition $\boldsymbol{u}(\boldsymbol{x}, \boldsymbol{s}) \in \mathcal{U} \otimes L^2(\Gamma_s)$

$$\boldsymbol{u}(\boldsymbol{x}, \boldsymbol{s}) \simeq \boldsymbol{u}^N = \sum_{m=1}^{N} \boldsymbol{\varphi}^m(\boldsymbol{x}) \circ \boldsymbol{\lambda}^m(\boldsymbol{s}) \tag{2}$$

4. Numerical results

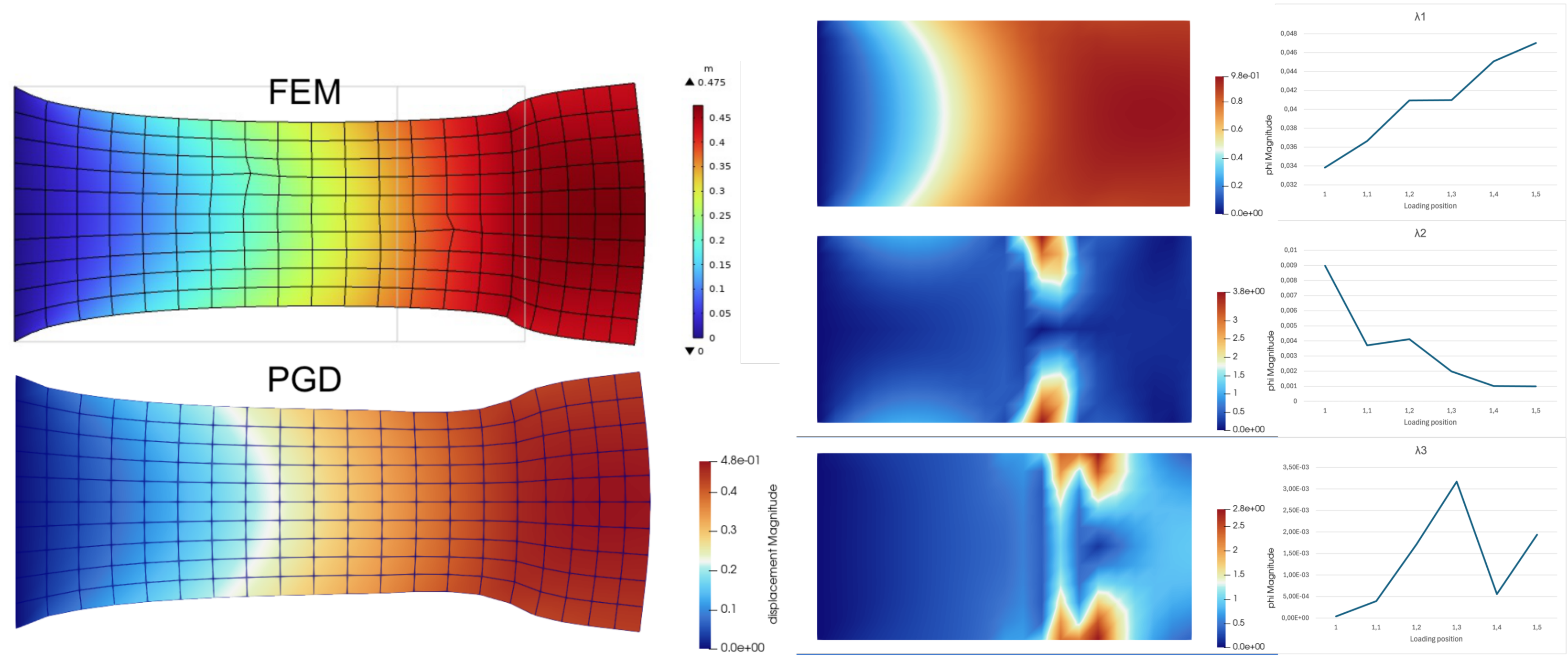


Figure 4: order 1, $n_{inc} = 10$, $N_{mod} = 3$: comparison between classic FEM and PGD solution (left) – the modes of the PGD solution for step $i = 10$ (right)

$$err_{x,y} = \frac{\|u^{PGD}_{x,y} - u^{FEM}_{x,y}\|_{\Omega\times\Gamma_s}}{\|\mathbf{u}^{FEM}\|_{\Omega\times\Gamma_s}} \tag{5}$$

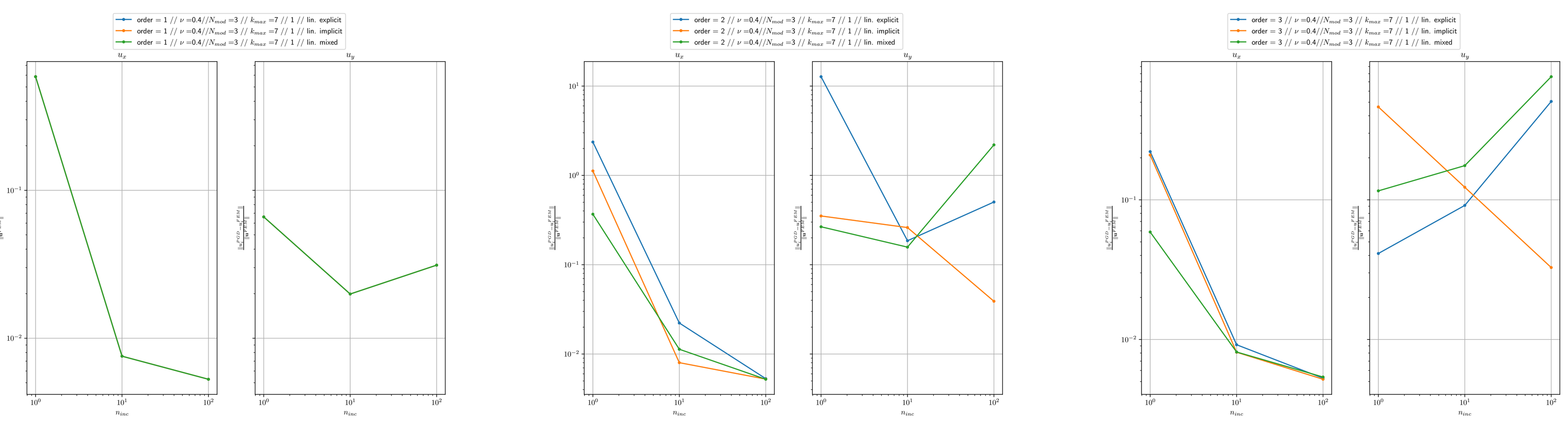


Figure 5: Errors for $N_{mod} = 3$, $n_{inc} = 10$: order 1 (left), 2 (middle), and 3 (right)

→ Encouraging results for SVK-I, but order 1 is better than 2 & 3.
→ Some investigating remains for SVK-M, which should behave closer to SVK-I.

References

Some more results